Towards Training Robustness Against Dynamic Errors in Quantum Machine Learning

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***Abstract*—Quantum machine learning, crucial in the noisy intermediate-scale quantum (NISQ) era, confronts challenges in error mitigation. Current noise-aware training (NAT) methods often assume static error rates in quantum neural networks (QNNs), overlooking the dynamic nature of quantum noise. Our work highlights how error rates fluctuate over time and across different qubits, affecting QNN performance even when overall error rates are similar. We introduce a novel NAT strategy that dynamically adjusts to standard and fatal error conditions, incorporating a low-complexity search method to identify fatal errors during optimization. This strategy significantly improves robustness, maintaining competitive performance with leading NAT methods across varying error scenarios.**

I. Introduction

Quantum machine learning (QML) is emerging in various practical applications [1]–[3]. Specifically, the variational quantum algorithm (VQA) [4] is promising to solve near-term QML tasks. The VQA is performed in the hybrid classical-quantum optimization scenario, where parameterized quantum circuits (PQCs), a.k.a. *ansatz*, produce measurements of quantum states on the quantum computer, and the classical computer performs parameter optimization of PQCs through gradient-based methods [5]. We demonstrate this procedure in Fig. 1, taking qauntum neural network (QNN) as an example.

However, the execution of PQCs faces rigorous situations with noise. As quantum computers move to the noisy intermediate-scale quantum (NISQ) era, errors in qubits and quantum operations are inevitable due to the dynamic nature of the quantum system and its environment [6]. Thus, error mitigation in PQCs is critical for robust execution on quantum computers, where mitigation strategies have been proposed during QNN training [7] or execution [8]–[10]. In this work, we focus on noise-aware training (NAT) for QNN models, aiming to enhance the robustness of PQC against errors. The current NAT approach proposed by [7] involves deliberately introducing simulated noise into quantum gates, based on a predefined error rate. Despite the pronounced effectiveness, we highlight a critical limitation in the current NAT strategy and PQC simulators, which rely on static error rates during noise analysis. ***In reality, the noise landscape is dynamic across both time and qubits.*** Recent study demonstrates that quantum noise levels can fluctuate significantly over time, even within a transient time window [11]. We further emphasize the dynamic nature of noise across different qubits. Such dynamic noise landscape renders the static error rate assumption in NAT insufficient for practice. For instance, a qubit in simulation

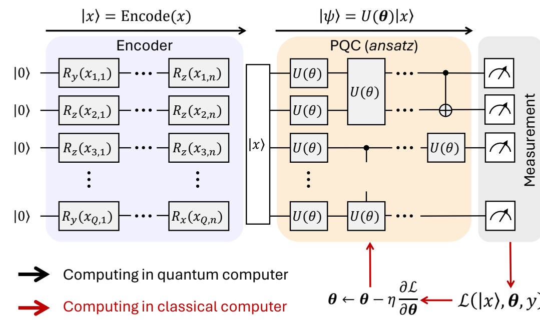

Fig. 1: A demonstration of QNN task. Input x is encoded as a state vector $|x\rangle$ using rotation gates [12], which is fed into PQC to prepare a final state vector $|\psi\rangle$ for measurement.

that is assumed to have a low error rate could be mapped to a physical qubit with a significantly higher error rate.

Additionally, current approaches to evaluating errors in noisy QNN models predominantly focus on the probability of errors, rather than assessing the actual detrimental impact these errors have on QNN performance. In practice, even errors with identical probabilities can have vastly different consequences. Certain errors may have a negligible influence on the QNN's inference performance, while some fatal errors can degrade its performance to levels worse than random guessing. The situation is more critical when qubits with possible fatal errors are mapped to high-error-rate physical qubits, posing a significant risk to the robustness of QNN inference.

In this paper, we characterize the influence of dynamic noise on PQCs, and demonstrate the scenarios where state-of-the-art (SOTA) NAT strategy is not competent. Our analysis underscores the need for a more comprehensive NAT strategy that transcends static error rate consideration and effectively mitigates fatal errors. To tackle this challenge, we propose a low-complexity search strategy. It can search for fatal errors along with QNN training, and evolve with PQC parameters simultaneously. Rather than randomly sampling errors, we utilize the searched fatal errors to boost the robust QNN training. We highlight our contributions as follows:

- We reveal the significant influence of dynamic noise landscape on QNN models. We also quantify the negative influence of error cases on QNN models, and propose the analysis methods for the fatal errors on a QNN model.
- To the best of our knowledge, this is the first NAT method

that is independent of error rates, but concentrates on fatal errors of PQCs. Our strategy aims to develop QNN models adaptive to various noise distributions and with good tolerance to these fatal error cases.

- We evaluate the effectiveness of our strategy on various QML tasks, which greatly improves the robustness of QNN models against fatal errors. Also, our strategy can achieve performance comparable to that of the SOTA NAT strategy under various error rates.

II. Preliminary

A. Quantum Neural Network with Noise

As illustrated in Fig. 1, a QNN model starts with a predefined encoder [7], [12], [13] to prepare the query x as a quantum state vector $|x\rangle$. It then applies a sequence of parameterized unitary gates $U_g(\theta_g) \in \mathcal{G}$ represented by unitary matrix $U(\boldsymbol{\theta}) = \prod_{g=G}^{1} U_g(\theta_g)$. The final quantum state is denoted as $|\psi\rangle = U(\boldsymbol{\theta})\,|x\rangle$. Then, the measurement of PQC is tackled by evaluating an observable B on $|\psi\rangle$, which is expressed as

$$f(|x\rangle, \boldsymbol{\theta}) = \langle\psi|\, B\, |\psi\rangle = \langle x|\, U^{\dagger}(\boldsymbol{\theta}) B U(\boldsymbol{\theta})\, |x\rangle \tag{1}$$

In the following, we use $f(|x\rangle, \boldsymbol{\theta})$ to denote the expected measurement results of PQC.

Noise Model. To simulate noise effects in PQCs, we consider quantum errors such as depolarizing noise and phase damping. They can be generally represented as `Pauli-X`, `Pauli-Y`, and `Pauli-Z` operation after gates, as outlined in [7]. Developing QNNs with inherent resilience to Pauli errors during training can significantly alleviate extensive error correction during execution. Additionally, post-training error mitigation methods, such as measurement error mitigation [9], have been widely explored and orthogonal to our approach. This combination is further explored in Sec. V-A.

Deterministic Modeling. The deterministic noise model estimates the PQC output in certain error cases [14]. Assuming an error E_g (as an operation) could occur on the gate $U_g(\theta_g)$, e.g., $U_g(\theta_g|E_g) = E_g U_g(\theta_g)$, we define the measurement as

$$f(|x\rangle, \boldsymbol{\theta}|\boldsymbol{E}) = \langle\psi|\, B\, |\psi\rangle\,, \text{ where } |\psi\rangle = U(\boldsymbol{\theta}|\boldsymbol{E})\,|x\rangle \tag{2}$$

The noisy PQC conducts $U(\boldsymbol{\theta}|\boldsymbol{E}) = \prod_{g=G}^{1} E_g U_g(\theta_g)$ on $|x\rangle$, where $\boldsymbol{E} \in \{I, X, Y, Z\}^G$ is a specific error case. Deterministic modeling tackles the analysis of error effect under certain errors, such as systemetic errors [15] and coherent errors [16]. In our work, we utilize this model to evaluate the worst-case errors for PQC measurement.

B. Related Work

The noise-aware training (NAT) on QNNs has been recently proposed to apply robust training strategies during parameter optimization [7]. Specifically, random noise injection is utilized during training, and other strategies, such as normalization and quantization, are applied as post-training robustness enhancement. Our work aligns with the in-training methodology, for example, noise injection, while it is orthogonal to the other strategies. The ensemble strategy is also proposed to improve performance with majority voting [17]; however, the multiplied

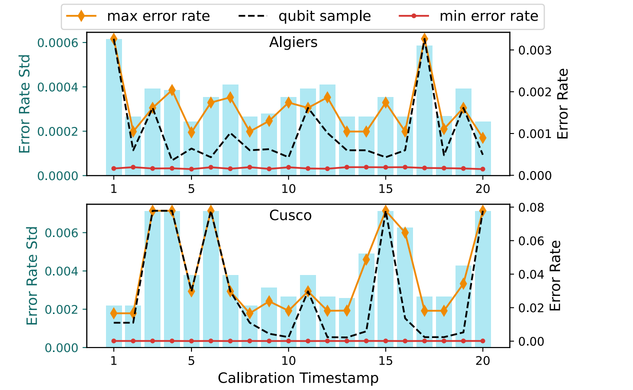

Fig. 2: Qubit error rates on both qubit scale (left y-axis) and time scale (right y-axis). We record the `Pauli-X` gate error rates of all qubits of two IBM quantum computers, `ibmq_algiers` (27 qubits) and `ibmq_cusco` (127 qubits).

model size will either burden the quantum resources or lengthen the execution time along the ensemble scale. Nevertheless, both works assume a static and unique noise distribution during QNN training, which contradicts reality, as quantum noise exhibits dynamic properties.

C. Scope of Our Work

Application Scenario. We aim to mitigate the dynamic error effect on quantum computers by training a robust model for *future use*. It should perform well under dynamic and unpredictable noise. Therefore, QNN models pre-trained through classical-quantum hybrid algorithms are ideal for this scenario. Other VQA tasks, such as VQE [18], fall outside our scope. For instance, once the state is optimized and prepared in VQE, no further execution of the PQC is needed. In such tasks, real-time error mitigation methods, including surface codes [19] and iterative skipping strategies [11], are more appropriate.

Case Study. In the following analysis, validation is provided as a numerical proof of concept along with the analytical dicussion. Without loss of generality, we take a case study from a 4-qubit QNN, whose PQC contains 3 layers. Each layer is composed of 4 `U3` gates and 4 `CU3` gates in a cyclical manner [12]. This model is trained for an MNIST-2 task that classifies digits 3 and 6.

III. Dynamic Noise and Fatal Error on QNN

A. Severe Error Variation on Time Scale and Qubit Scale

In previous NAT on QNN, the model parameters are usually optimized under the static noise assumption. However, transient errors have recently been recognized for both long-term and short-term quantum operations [11], where the characteristic of qubits could vary significantly along the ***time*** scale.

Even worse, large variations of qubit characteristics also exist along the ***qubit*** scale. In Fig. 2, we demonstrate the dynamic noise landscape from two IBM commercial quantum computers, on the time scale and the qubit scale. During a long-term run, the gate error rate on a certain qubit (e.g., shown as

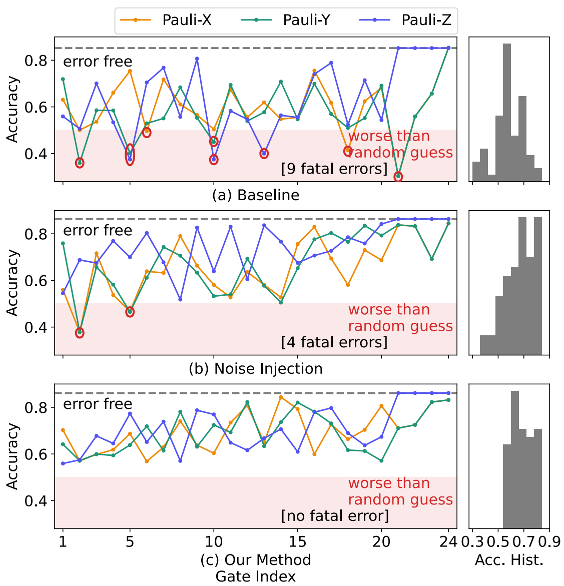

Fig. 3: The inference accuracy of QNN, trained by (a) no error consideration (b) noise-injection strategy and (c) our method (Sec. IV), for the MNIST-2 task, assuming only one gate occurring logical error. Right plots are the corresponding histograms of accuracies.

black dash lines in Fig. 2) could vary by an order of magnitude in the worst case. On the qubit scale, the error rates of different qubits also have a significant variation. In some extreme cases, the error rate on one qubit can be 80 times greater than that on another. Thus, for a well-trained QNN, different mappings from its qubits to the physical qubits on quantum computers can induce significant error rate divergence on each qubit.

B. Fatal Errors Poisoning QNN Accuracy

Besides the error rate variation, not all gates in a PQC have the same sensitivity to errors. We present this assessment of QNN in Fig. 3, under a minimal error scenario where only one single gate is affected by error. The result is intriguing, since even one gate error can significantly compromise the QNN model into a useless level. Specifically, there are 9 out of 72 errors, for a QNN trained without noise consideration, that result in performance below random guess. Even with SOTA noise injection training [7], there are still 4 errors that lead to worse than random guess. This underscores that errors at different locations can cause widely disparate levels of performance degradation, even with similar error rates. On real quantum computers, the occurrence of such a fatal error can render the QNN inference utterly unreliable. Under the dynamic noise, the coming of fatal error cases is also unpredictable.

As a preview, our strategy significantly enhances QNN performance across all error cases, ensuring that predictions consistently exceed random guesses in all scenarios explored in our case study. Furthermore, accuracy histograms indicate that SOTA noise injection training generally improves model accuracy under most error conditions over the error-free training (baseline). However, this approach falls short in addressing specific fatal error scenarios effectively. In contrast, our method successfully mitigates the impact of these fatal errors without compromising model performance in other error scenarios.

C. Worst Case Analysis

Evaluating all possible error scenarios during QNN training is both time-consuming and impractical, especially since many of these scenarios have negligible probabilities. Instead, it is more meaningful to focus on error cases with relatively high probabilities, as these are the most common during quantum execution. For example, with an average error rate of 0.001 (such as in `ibmq_cusco`) across a 100-gate PQC, the conditional probability that one or two errors occur given the error occurring, is $P(N_{\mathcal{E}} \leq 2 | N_{\mathcal{E}} \geq 1) \approx 0.9984$ in normal use. Therefore, we emphasize concentrating on these "high-probability" errors, since they are the most likely to affect the model's performance when errors occur. In our analysis, we locate fatal errors in the high-probability error set and utilize the deterministic noise model described in Eq. 2.

Fatal loss $\mathcal{L}_{sup}$. Given an error set $\Omega_{\boldsymbol{e}}$ and input-label pair (x, y), we define the fatal loss to $f(|x\rangle, \boldsymbol{\theta})$, as the supremum of losses caused by all $\boldsymbol{E} \in \Omega_{\boldsymbol{e}}$,

$$\mathcal{L}_{sup}(\Omega_{\boldsymbol{e}}, x, y) = \sup_{\boldsymbol{E} \in \Omega_{\boldsymbol{e}}} \{\mathcal{L}\left(f(|x\rangle, \boldsymbol{\theta}|\boldsymbol{E}), y\right)\} \tag{3}$$

Accordingly, the QNN fatal accuracy evaluates the worst-case performance of PQC measurement against all noise cases in $\Omega_{\boldsymbol{e}}$. Our metric in Eq. 3 does not involve gate error rates during evaluation; thus, the evaluation remains fair and robust against dynamic noise changes in quantum computers.

Error Set Collection. The "high-probability" $\Omega_{\boldsymbol{e}}$ is collected from typical error cases observed under normal operating conditions, given the scale of the PQC and the quantum system layout. For instance, as aforementioned, it is reasonable to assume that the execution of a PQC on `ibmq_cusco` involves no more than three gate errors. Extreme scenarios with higher numbers of gate errors can be disregarded, as large fluctuations with very low probabilities are effectively averaged out during execution shots without introducing significant bias.

IV. Methodology: Equitable Training with Low-Complexity Search

A. Problem Formulation

We aim to co-optimize the QNN accuracy under general noises and fatal error cases, given error set $\Omega_{\boldsymbol{e}}$. We specifically consider fatal accuracy by actively searching for and sampling the fatal errors during optimization. In this error-rate-independent optimization problem, we do not rely on specific error rates; therefore, we treat the first component as an error-free term. The optimization problem is formulated as follows:

$$\min_{\boldsymbol{\theta}} \quad \mathcal{L}\left(f(|\mathbf{X}\rangle, \boldsymbol{\theta}), \mathbf{Y}\right) + \lambda \mathcal{L}_{sup}(\Omega_{\boldsymbol{e}}, \mathbf{X}, \mathbf{Y}) \tag{4}$$

where $(\mathbf{X}, \mathbf{Y})$ is the input-label pairs following distribution in sample space $\mathcal{X} \times \mathcal{Y}$, and $\lambda > 0$ is a hyperparameter to balance the focus on error-free accuracy and fatal accuracy.

TABLE I: Estimated $\mathcal{L}_{sup}(\Omega_{\boldsymbol{e}})$ on a data batch and time consumption on different strategies. The first column is the QNN scale and the maximal expected gate errors in $\Omega_{\boldsymbol{e}}$, i.e., (G, M). Search time is measured on Intel i7 @3.80GHz CPU.

Search Size	BruteForce		RandomSearch		**Low-Complexity Search**	
	$\mathcal{L}_{sup}(\Omega_{\boldsymbol{e}})$	Time(s)	$\widehat{\mathcal{L}}_{sup}(\Omega_{\boldsymbol{e}})$	Time(s)	$\widehat{\mathcal{L}}_{sup}(\Omega_{\boldsymbol{e}})$	Time(s)
$(24, 2)$	1.19	30	1.07	1.73	1.15	1.22
$(40, 2)$	1.11	122	1.03	4.24	1.06	1.74
$(56, 2)$	1.22	317	1.06	7.37	1.22	2.26
$(24, 3)$	1.31	655	1.12	2.57	1.27	1.20
$(40, 3)$	1.24	4,617	0.97	6.24	1.11	1.75
$(56, 3)$	1.23	15,964	1.09	10.40	1.22	2.35

B. Low-Complexity Search

To determine the supremum of losses, one can naively traverse all error cases in $\Omega_{\boldsymbol{e}}$. However, the cardinality of the error set $|\Omega_{\boldsymbol{e}}|$ actually increases polynomially along the PQC scale, i.e., the gate number G. Since $\mathcal{L}_{sup}$ must be evaluated for each data batch and at every iteration during training, the computational time quickly becomes impractical as the PQC scales up. Therefore, it is essential to adopt low-complexity yet effective search strategies to address this scalability challenge.

Scalable Evolution-Based Low-Complexity Search. We adopt the idea from genetic algorithm [20] to search for fatal errors during each training iteration. To balance the search effort and effectiveness, we propose allowing the error cases to evolve alongside the model optimization. Under this strategy, the search complexity in each iteration only depends on the error population N_{pop} in current generation.

Low-Complexity Search Validation in Fatal Loss Estimation. The evaluation of $\mathcal{L}_{sup}(\Omega_{\boldsymbol{e}})$ must be performed repeatedly during QNN training, thus the key to reducing time consumption is to use fewer samples to derive a highly accurate estimate. In Table I, we demonstrate the search procedure between brute-force search, random search, and our low-complexity search. We construct three QNNs with scales ($G = 24, 40, 56$) on the MNIST-2 task, and vary $\Omega_{\boldsymbol{e}}$ by collecting all cases up to two ($M = 2$) and three ($M = 3$) errors. While brute-force search explores all error cases in $\Omega_{\boldsymbol{e}}$, it guarantees an accurate fatal loss estimate, but at the cost of significant computational effort. Random search (utilized in [7]), though less exhaustive, fails to provide sufficiently accurate estimates of fatal loss. Our evolution-based low-complexity search strategy offers comparable estimates to sequential search but with approximately half the time cost for small-to-moderate QNN scales; this speedup even increases over 4 times on large QNN. Moreover, the time cost of our low-complexity search grows only marginally with the number of errors, e.g., only around 4% from two errors to three errors, while this increment is significant on the other two strategies.

C. Overview of Equitable Training

To solve the optimization problem presented in Eq. 4, we iteratively update the parameters $\boldsymbol{\theta}$ ithrough equitable training, as detailed in Algorithm 1. In each iteration, we perform signle-generation low-complexity search to estimate the supremum of $\mathcal{L}_{sup}(\Omega_{\boldsymbol{e}})$ for each data batch, so that error candidates evolve

Algorithm 1 Equitable noise-aware training on QNN

Require: error case set Ω_e, QNN model f, training epoch T, scaling factor λ, learning rate η

Ensure: $\boldsymbol{\theta}^{\star} = \boldsymbol{\theta}^{T}$ for equitable QNN

```
1: initialize θ^0 and candidates S_e ⊂ Ω_e;
2: for t = 1..T do
3:     for data batch x do
4:         E*, {L_E} ← sup_{E∈S_e} L(f(|x⟩, θ^{t-1}|E), y)
                                    ▷ Deterministic noise model
5:         S_e ⊂ Ω_e ← crossover&mutation(S_e, {L_E})
6:         ∂L/∂θ, ∂L'/∂θ ← ∂L(f(|x⟩,θ^{t-1}),y)/∂θ^{t-1}, ∂L(f(|x⟩,θ^{t-1}|E*),y)/∂θ^{t-1}
7:         θ^t ← θ^{t-1} − η ∂L/∂θ − ηλ ∂L'/∂θ
8:     end for
9: end for
```

with the QNN training. This is followed by evaluating both the error-free model and the noisy model with identified fatal error $\boldsymbol{E}^{\star}$. Crucially, we conduct multiple searches per iteration to frequently refresh the candidate set $\mathcal{S}_{\boldsymbol{e}}$. This approach is essential for enhancing the robustness of the QNN model against critical errors, given the large size of $\Omega_{\boldsymbol{e}}$ and the necessity of covering fatal error cases for effective mitigation.

λ **Selection.** λ provides the trade-off between the normal accuracy and fatal accuracy of the QNN models. Although the determination of λ can be evaluated by hyperparameter tuning, we share an empirical start as $\lambda = 0.5$.

V. Evaluation

Benchmark Strategies. All QNN tasks are evaluated under error-free training (baseline) and noise injection training (NI) strategies, as benchmark methods. For baseline, we optimize QNN models excluding noise consideration during QNN training. For NI, we follow the assumption in [7] that one gate has the same static rates to induce `Pauli-X`, `Pauli-Y`, and `Pauli-Z` error. For the injecting error rate, we generally categorize the error rates of a QNN model as "NI-L(ight)" ($p^X = p^Y = p^Z = 0.001$ for all qubits), "NI-M(oderate)" ($p = 0.005$), and "NI-H(eavy)" ($p = 0.01$).

Metric. We apply fatal accuracy (F.Acc) and noisy accuracy (N.Acc) to evaluate different NAT strategies. Specifically, fatal accuracy is derived from the lowest accuracy from error cases of $\Omega_{\boldsymbol{e}}$ that contain high-probability errors. On the other hand, the noisy accuracy measures the model performance under a certain set of error rates on qubits. As the qubit error rate is unpredictable and dispersed (Section III)[1], we randomly pick three noisy environments based on the error rate levels. We denote $p_g = p_g^X = p_g^Y = p_g^Z$ on g-th gate, and each gate's error rate follows Gaussian distribution $p_g \sim \mathcal{N}(\mu_p, \sigma_p^2)$ s.t. $p_g > 0$ for a G-gate PQC. N.Acc(L) – the noisy accuracy under low error rates, $\mu_p = 0.001$ and $\sigma_p = 0.0005$; N.Acc(M) – for medium error rates $\mu_p = 0.005$ and $\sigma_p = 0.002$; N.Acc(H) – for high error rates $\mu_p = 0.01$ and $\sigma_p = 0.005$.

[1]This is why we avoid using real quantum environments, as their error rates cannot be accurately quantified or customized.

TABLE II: The fatal accuracy (F.Acc) on QNN models (with 3/5/7/10 layers), trained by the baseline, noise-injection, and ours strategy. The evaluation tasks are MNIST (M2, M4, M10), FMNIST(F2, F4), and CIFAR(C2). All the models are trained on 50 epochs, with the same training configuration. Best F.Acc for each task is **highlighted**. Additionally, improvements of ours over the second-best are marked in green, while the difference to the best is in red when ours is not the best.

	3Layer						5Layer				7Layer				10Layer			
	M2	M4	M10	F2	F4	C2	M2	M10	F2	F4	M2	M10	F2	C2	M2	F2	F4	C2
Baseline	0.303	0.023	0.067	0.247	0.099	0.380	0.152	0.025	0.163	0.059	0.146	0.011	0.166	0.378	0.141	**0.195**	0.056	0.341
NI-L	0.378	0.047	0.094	0.293	0.141	0.453	0.152	0.025	0.165	0.055	0.136	0.027	0.152	0.387	0.135	0.160	0.055	0.396
NI-M	0.273	0.033	0.062	0.333	0.147	0.444	0.148	0.023	0.164	0.041	0.149	0.030	0.167	0.388	0.127	0.164	0.039	0.367
NI-H	0.390	0.059	0.100	0.498	0.108	0.501	0.179	0.042	0.184	0.044	0.152	**0.089**	0.164	0.368	0.174	0.171	0.052	**0.397**
ours	**0.564**	**0.136**	**0.139**	**0.673**	**0.272**	**0.536**	**0.184**	**0.074**	**0.211**	**0.064**	**0.191**	0.086	**0.198**	**0.407**	**0.183**	0.173	**0.057**	0.390
(+/−)	+0.174	+0.077	+0.039	+0.175	+0.126	+0.036	+0.004	+0.032	+0.027	+0.005	+0.039	-0.003	+0.031	+0.019	+0.009	-0.022	+0.001	-0.007

A. Problem Solving: Image Classification

As image classification on QNN is still emerging, common QNN models can achieve good performance on easy datasets, such as MNIST [21], FMNIST [22], and CIFAR [23]. We evaluate the tasks MNIST2, MNIST4, MNIST10, FMNIST2, FMNIST4, and CIFAR2, which all have four-qubit input except MNIST10 with ten qubits.

QNN Configuration. W.L.O.G, we build the QNN with `U3` gates and `CU3` (Controlled-U3) gates. We define one layer as "`U3+CU3`" layer, where each qubit has a `U3` gate followed by a `CU3` gate cyclically [12]. We specify four QNN models, with 3 layers, 5 layers, 7 layers, and 10 layers. To apply our equitable training, we empirically assume $M = 1, 2, 2,$ and 3 for each QNN model, representing larger PQC scales with more expected gate errors in practice.

Fatal Accuracy Evaluation. We demonstrate the fatal accuracies of different training strategies on our selected QNN models in Table II. From an overall standpoint, our equitable training strategy consistently achieves the highest fatal accuracy across most tasks. It only slightly underperforms compared to the top-performing strategies on some large-scale tasks, such as the 7-layer MNIST10, where the difference is minimal. By comparing the NI strategy with the baseline method, we only observe a marked improvement in fatal accuracy in smaller QNNs (e.g., 3 layers). However, the advantage of the NI strategy diminishes for moderate to large QNNs (5 layers and above). This is evidenced by the NI strategy's marginal performance improvement over the baseline in fatal accuracy assessments. Only the "NI-Heavy" method, which has a higher likelihood of sampling errors, occasionally outperforms the baseline method with certain improvements.

On the other hand, our equitable training strategy significantly outperforms the NI strategy in small-scale QNNs, e.g., achieving a 17.51% improvement in FMNIST2 for 3-layer QNN. Due to more frequent access of fatal errors during training, our strategy also excels in moderate-scale QNN models, such as those with 5 layers. Yet, as the model scale increases (in 10-layer QNNs), our advantage diminishes, indicating reduced access to fatal errors and less optimization of the PQC parameters regarding these errors. Despite these challenges, our strategy still manages to deliver the best performance in over half of the large-scale QNN tasks.

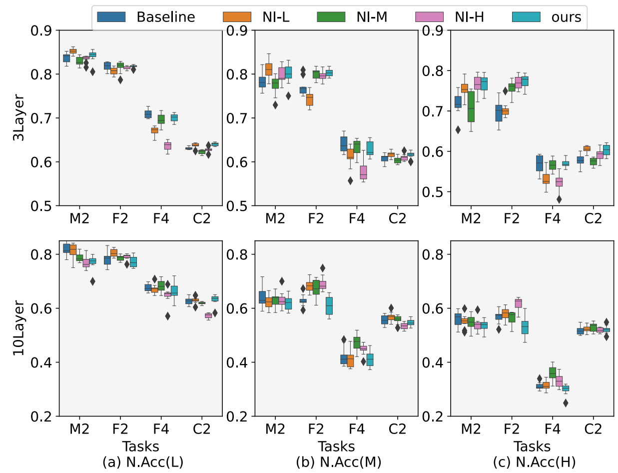

Fig. 4: The noisy accuracy (N.Acc) of the 3/10-layer QNN model, optimized using various strategies under (L)ow, (M)edium, and (H)igh error rates.

Noisy Accuracy Evaluation. In Fig. 4, we demonstrate the noisy accuracy of the small (3-layer) and large (10-layer) QNN models in different noisy environments. By incorporating additional parameters into PQC, the 10-layer model exhibits a higher likelihood of errors in noisy environments than 3 layers, leading to an obvious decline in noisy accuracy. Notably, additional layers would typically enhance performance without noise, creating antagonsim with the detrimental effects of more errors in the noisy environment. This is demonstrated by the evaluation for CIFAR2 task across 3-layer to 10-layer QNN, where the 10-layer QNN surpasses the 3-layer model under low error conditions (N.Acc(L)), yet accuracy decreases with further layer additions. In most cases, the adverse impact of errors outweighs the benefits of increased model complexity. Therefore, effective error mitigation strategies are essential for practical QML tasks.

Analysis of different strategies reveals that the baseline method excels only in simpler tasks, such as MNIST2, while strategies that incorporate noise considerations generally outperform the baseline in noisy environments across other tasks. However, this advantage is not markedly pronounced. Moreover, when compared to NI strategies, our equitable training achieves comparable noisy accuracy, leading in 8 out of 24 tasks.

TABLE III: The N.Acc of 3-layer QNN under different noise mitigation strategies, under noisy environment of $p = 0.01$. Accuracies are collected from 10 runs. EX.—Extrapolation.

	M2	M4	F2	F4
Baseline	$0.718^{\pm 0.015}$	$0.488^{\pm 0.023}$	$0.710^{\pm 0.014}$	$0.576^{\pm 0.010}$
ours	$0.763^{\pm 0.009}$	$0.520^{\pm 0.014}$	$0.774^{\pm 0.007}$	$0.562^{\pm 0.012}$
ours+EX.	$0.811^{\pm 0.015}$	$0.639^{\pm 0.008}$	$0.802^{\pm 0.007}$	$0.642^{\pm 0.010}$
ours+Norm.	$0.766^{\pm 0.026}$	$0.567^{\pm 0.022}$	$0.772^{\pm 0.032}$	$0.616^{\pm 0.022}$
error-free	0.856	0.691	0.822	0.715

TABLE IV: Fatal accuracy (F.Acc) and noisy accuracy(N.Acc) for the POS tagging task under different NAT strategies. The noisy accuracy is averaged on 10 runs.

	F.Acc	N.Acc(L)	N.Acc(M)	N.Acc(H)
Baseline	0.35	$0.92^{\pm 0.02}$	$0.90^{\pm 0.02}$	$0.87^{\pm 0.03}$
NI-L	0.18	$0.90^{\pm 0.01}$	$0.87^{\pm 0.03}$	$0.85^{\pm 0.04}$
NI-M	0.20	$0.94^{\pm 0.01}$	$0.92^{\pm 0.02}$	$0.89^{\pm 0.04}$
NI-H	0.13	$0.89^{\pm 0.01}$	$0.88^{\pm 0.03}$	$0.85^{\pm 0.05}$
Ours	**0.69**	$\mathbf{0.94}^{\pm 0.01}$	$\mathbf{0.93}^{\pm 0.01}$	$\mathbf{0.90}^{\pm 0.04}$

Combined with our method's superior performance in fatal accuracy, our equitable training not only matches the general performance of SOTA NAT strategies without compromise, but also significantly enhances the robustness of the QNN models against fatal errors during practical execution.

NAT with Noise Mitigation. We assess the integration of our NAT strategy with post-training noise mitigation techniques in Table III. For the extrapolation method [24], we selecte 10 random error rates between 0.001 and 0.05 during QNN inference and extrapolate to the error-free expectation value of the PQC measurement. In measurement normalization [7], the QNN model is trained to normalize each qubit measurement across data batches. As our training strategy imporves N.Acc on most tasks compared to the baseline, post-training noise mitigation further improves inference accuracy, approximating error-free conditions. Among the strategies, extrapolation yield more substantial improvements due to multiple measurements, whereas measurement normalization provides enhancements with a single inference run.

B. Problem Solving: POS Tagging

We take a case study of the part-of-speech (POS) tagging task to evaluate the NAT strategies in another QNN architecture, i.e., quantum long short-term memory (QLSTM) [25]. Given that current QLSTM models can only be performed on small-scale datasets [26], popular ones (such as Penn Treebank [27]) are not practical for QLSTM application, and to our knowledge there is no corresponding work on them. Thus, we generate a small corpus that contains 20 sentences, whose vocabulary tags contain noun, verb, determiner, adjective, adverb, and pronoun.

Results. In Table IV, we evaluate the fatal accuracy and noisy accuracy on QNN models optimized with different strategies. For the fatal accuracy, our training strategy can achieve a significant improvement over baseline and noise-injection methods. Since the QNN scale and the cardinality of Ω_e are small and we have enough search times during the training iteration, the fatal loss is well addressed. Besides, since the QLSTM architecture is a hybrid model, i.e., classical NN plus QNN, the parameters in the classical part can also help the robustness of QNN against fatal errors. In addition, our method can achieve comparable noisy accuracies with the noise-injection strategy under various error rates. This observation again proves the superiority of our method to provide robustness in fatal errors and dynamic noise environments.

TABLE V: The fatal MSE loss (F.Loss) and noisy MSE loss (N.Loss) with different error rates, for the regression task under different NAT strategies. The noisy loss is averaged on 10 runs.

	F.Loss	N.Loss(L)	N.Loss(M)	N.Loss(H)
Baseline	1.958	$\mathbf{0.008}^{\pm 0.025}$	$0.075^{\pm 0.102}$	$0.404^{\pm 0.279}$
NI-L	1.261	$0.029^{\pm 0.058}$	$0.076^{\pm 0.195}$	$0.245^{\pm 0.156}$
NI-M	1.695	$0.055^{\pm 0.096}$	$0.059^{\pm 0.073}$	$0.503^{\pm 0.249}$
NI-H	1.292	$0.010^{\pm 0.026}$	$\mathbf{0.054}^{\pm 0.070}$	$0.196^{\pm 0.172}$
Ours	**0.543**	$0.048^{\pm 0.019}$	$0.123^{\pm 0.083}$	$\mathbf{0.134}^{\pm 0.088}$

C. Problem Solving: Regression

QNNs have been explored in regression tasks [28]. We build the QNN as a 3-layer `U3+CU3` model with 2 qubits, which has in total $G = 12$ gates, thus the Ω_e collects all the error cases where $M = 1$. We evaluate the regression task to fit the mapping function: $y = \sin(2x_1)\cos(x_2)$.

Results. Table V shows the fatal and noisy losses on training strategies evaluated. Our equitable training can achieve at least 57% reduction on the fatal loss over other strategies. For noisy loss, the divergence of performance of different methods is significant. The baseline strategy without error consideration is prone to have lower loss in low error rate environment, while the noise injection with high-error-rate (NI-H) and our method perform better in high error rate environment. In addition, we notice that the noisy loss evaluation is not accurate on large error rate, w.r.t. large standard deviation to the average N.Loss. This is also a motivation that we propose the F.Loss metric to evaluate the robustness of a model against error, which is error-rate independent.

VI. Conclusion

We highlight the critical impact of dynamic noise and fatal errors on QNNs. Existing training and evaluation methods fall short facing a dynamically changing quantum noise environment. We propose fatal error analysis, which is independent of error rates and assesses QNN robustness against high-probability fatal errors. We also present the first noise-aware training strategy aimed at general fatal error mitigation. Our evaluations confirm the efficacy of our strategy for fatal errors; besides, QNNs optimized by our approach perform on par with SOTA NAT strategies under various error rates.

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REFERENCES

[1] M. Sajjan, J. Li, R. Selvarajan, S. H. Sureshbabu, S. S. Kale, R. Gupta, V. Singh, and S. Kais, "Quantum machine learning for chemistry and physics," *Chemical Society Reviews*, 2022.

[2] W. Guan, G. Perdue, A. Pesah, M. Schuld, K. Terashi, S. Vallecorsa, and J.-R. Vlimant, "Quantum machine learning in high energy physics," *Machine Learning: Science and Technology*, vol. 2, no. 1, p. 011003, 2021.

[3] R. Guarasci, G. De Pietro, and M. Esposito, "Quantum natural language processing: Challenges and opportunities," *Applied Sciences*, vol. 12, no. 11, p. 5651, 2022.

[4] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio *et al.*, "Variational quantum algorithms," *Nature Reviews Physics*, vol. 3, no. 9, pp. 625–644, 2021.

[5] O. Simeone *et al.*, "An introduction to quantum machine learning for engineers," *Foundations and Trends® in Signal Processing*, vol. 16, no. 1-2, pp. 1–223, 2022.

[6] J. Preskill, "Quantum computing in the nisq era and beyond," *Quantum*, vol. 2, p. 79, 2018.

[7] H. Wang, J. Gu, Y. Ding, Z. Li, F. T. Chong, D. Z. Pan, and S. Han, "Quantumnat: quantum noise-aware training with noise injection, quantization and normalization," in *Proceedings of the 59th ACM/IEEE Design Automation Conference*, 2022, pp. 1–6.

[8] Z. Cai, R. Babbush, S. C. Benjamin, S. Endo, W. J. Huggins, Y. Li, J. R. McClean, and T. E. O'Brien, "Quantum error mitigation," *Reviews of Modern Physics*, vol. 95, no. 4, p. 045005, 2023.

[9] S. Endo, Z. Cai, S. C. Benjamin, and X. Yuan, "Hybrid quantum-classical algorithms and quantum error mitigation," *Journal of the Physical Society of Japan*, vol. 90, no. 3, p. 032001, 2021.

[10] A. Strikis, D. Qin, Y. Chen, S. C. Benjamin, and Y. Li, "Learning-based quantum error mitigation," *PRX Quantum*, vol. 2, no. 4, p. 040330, 2021.

[11] G. S. Ravi, K. Smith, J. M. Baker, T. Kannan, N. Earnest, A. Javadi-Abhari, H. Hoffmann, and F. T. Chong, "Navigating the dynamic noise landscape of variational quantum algorithms with qismet," in *Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2*, 2023, pp. 515–529.

[12] H. Wang, Y. Ding, J. Gu, Y. Lin, D. Z. Pan, F. T. Chong, and S. Han, "Quantumnas: Noise-adaptive search for robust quantum circuits," in *2022 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*. IEEE, 2022, pp. 692–708.

[13] F. Yan, A. M. Iliyasu, and S. E. Venegas-Andraca, "A survey of quantum image representations," *Quantum Information Processing*, vol. 15, pp. 1–35, 2016.

[14] M. A. Nielsen and I. L. Chuang, *Quantum computation and quantum information*. Cambridge university press, 2010.

[15] K. Khodjasteh and L. Viola, "Dynamically error-corrected gates for universal quantum computation," *Physical review letters*, vol. 102, no. 8, p. 080501, 2009.

[16] D. Greenbaum and Z. Dutton, "Modeling coherent errors in quantum error correction," *Quantum Science and Technology*, vol. 3, no. 1, p. 015007, 2017.

[17] R. Qin, Z. Liang, J. Cheng, P. Kogge, and Y. Shi, "Improving quantum classifier performance in nisq computers by voting strategy from ensemble learning," *arXiv preprint arXiv:2210.01656*, 2022.

[18] J. Tilly, H. Chen, S. Cao, D. Picozzi, K. Setia, Y. Li, E. Grant, L. Wossnig, I. Rungger, G. H. Booth *et al.*, "The variational quantum eigensolver: a review of methods and best practices," *Physics Reports*, vol. 986, pp. 1–128, 2022.

[19] R. Acharya *et al.*, "Suppressing quantum errors by scaling a surface code logical qubit," *Nature*, vol. 614, no. 7949, pp. 676–681, 2023.

[20] S. Mirjalili and S. Mirjalili, "Genetic algorithm," *Evolutionary algorithms and neural networks: theory and applications*, pp. 43–55, 2019.

[21] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[22] H. Xiao, K. Rasul, and R. Vollgraf. (2017) Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms.

[23] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," 2009.

[24] K. Temme, S. Bravyi, and J. M. Gambetta, "Error mitigation for short-depth quantum circuits," *Physical review letters*, vol. 119, no. 18, p. 180509, 2017.

[25] S. Y.-C. Chen, S. Yoo, and Y.-L. L. Fang, "Quantum long short-term memory," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 8622–8626.

[26] R. Di Sipio, J.-H. Huang, S. Y.-C. Chen, S. Mangini, and M. Worring, "The dawn of quantum natural language processing," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 8612–8616.

[27] M. Marcus, B. Santorini, and M. A. Marcinkiewicz, "Building a large annotated corpus of english: The penn treebank," 1993.

[28] G. Wang, "Quantum algorithm for linear regression," *Physical review A*, vol. 96, no. 1, p. 012335, 2017.